MAT 2040 Linear Algebra

Assignment 6

- Released date:
- Due:
- Late submission is NOT acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW6".

Question 1. For each of the following, compute (i) $\det(\boldsymbol{A})$, (ii) adj $\boldsymbol{A}$, and (iii) $\boldsymbol{A}^{-1}$:

$$\text{(a). } \boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \qquad \text{(b). } \boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution

(a). $\det(\boldsymbol{A}) = -7$, $\quad \text{adj } \boldsymbol{A} = \begin{bmatrix} -1 & -2 \\ -3 & 1 \end{bmatrix}$, $\quad \boldsymbol{A}^{-1} = -\frac{1}{7} \text{adj } \boldsymbol{A}$.

(b). $\det(\boldsymbol{A}) = 1$, $\quad \text{adj } \boldsymbol{A} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$, $\quad \boldsymbol{A}^{-1} = \text{adj } \boldsymbol{A}$.

Question 2. Use Cramer's rule to solve each of the following systems:

$$\text{(a). } \begin{cases} x_1 + 2x_2 = 3 \\ 3x_1 - x_2 = 1 \end{cases} . \qquad \text{(b). } \begin{cases} x_1 + 3x_2 + x_3 = 1 \\ 2x_1 + x_2 + x_3 = 5 \\ -2x_1 + 2x_2 - x_3 = -8 \end{cases} .$$

Solution

(a).

$$x_1 = \frac{\det(A_1)}{det(A)} = \frac{-5}{-7} = \frac{5}{7}$$
$$x_2 = \frac{\det(A_2)}{det(A)} = \frac{-8}{-7} = \frac{8}{7}$$

(b).

$$x_1 = \frac{\det(A_1)}{det(A)} = \frac{6}{3} = 2$$
$$x_2 = \frac{\det(A_2)}{det(A)} = \frac{-3}{3} = -1$$
$$x_3 = \frac{\det(A_3)}{det(A)} = \frac{6}{3} = 2$$

Question 3. Let

$$\boldsymbol{P} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}, \quad \boldsymbol{\Lambda} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{AP} = \boldsymbol{P\Lambda}.$$

Find $\boldsymbol{A}^n$.

Solution

$|\boldsymbol{P}| = 2, \boldsymbol{P}^{-1} = \frac{1}{2}\begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix}$,

$\boldsymbol{A}^n = (\boldsymbol{P\Lambda P}^{-1})(\boldsymbol{P\Lambda P}^{-1})(\boldsymbol{P\Lambda P}^{-1})\dots(\boldsymbol{P\Lambda P}^{-1})$
$= \boldsymbol{P\Lambda}(\boldsymbol{P}^{-1}\boldsymbol{P})\boldsymbol{\Lambda}(\boldsymbol{P}^{-1}\boldsymbol{P})\boldsymbol{\Lambda P}^{-1})\dots(\boldsymbol{P\Lambda P}^{-1}) = \boldsymbol{P\Lambda}^n\boldsymbol{P}^{-1}$

$$\boldsymbol{\Lambda} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \boldsymbol{\Lambda}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2^2 \end{bmatrix}, \dots, \boldsymbol{\Lambda}^n = \begin{bmatrix} 1 & 0 \\ 0 & 2^n \end{bmatrix},$$
$$\boldsymbol{A}^n = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 2^n \end{bmatrix}\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 2^{n+1} \\ 1 & 2^{n+2} \end{bmatrix}\begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix},$$
$$= \frac{1}{2}\begin{bmatrix} 4-2^{n+1} & 2^{n+1}-2 \\ 4-2^{n+2} & 2^{n+2}-2 \end{bmatrix} = \begin{bmatrix} 2-2^{\mathrm{n}} & 2^{\mathrm{n}}-1 \\ 2-2^{n+1} & 2^{n+1}-1 \end{bmatrix}.$$

Question 4. Show that if $\boldsymbol{A}$ is nonsingular, then $\mathrm{adj}(\boldsymbol{A})$ is nonsingular and

$$(\mathrm{adj}\,\boldsymbol{A})^{-1} = \det(\boldsymbol{A}^{-1})\boldsymbol{A} = \mathrm{adj}(\boldsymbol{A}^{-1}).$$

Solution

If $\boldsymbol{A}$ is nonsingular, then $\det(\boldsymbol{A}) \neq 0$ and hence

$$\operatorname{adj} \boldsymbol{A} = \det(\boldsymbol{A})\boldsymbol{A}^{-1}$$

is also nonsingular. It follows that

$$(\operatorname{adj} \boldsymbol{A})^{-1} = \frac{1}{\det(\boldsymbol{A})}(\boldsymbol{A}^{-1})^{-1} = \det(\boldsymbol{A}^{-1})\boldsymbol{A}.$$

Also

$$\operatorname{adj} \boldsymbol{A}^{-1} = \det(\boldsymbol{A}^{-1})(\boldsymbol{A}^{-1})^{-1} = \det(\boldsymbol{A}^{-1})\boldsymbol{A}.$$

Question 5. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be 3×3 matrices, with $\det(\boldsymbol{A}) = -3$ and $\det(\boldsymbol{B}) = -4$. Use the properties of determinants to compute:
(a). $\det(\boldsymbol{AB})$; (b). $\det(5\boldsymbol{A})$; (c). $\det(\boldsymbol{B}^T)$; (d). $\det(\boldsymbol{A}^{-1})$; (e). $\det(\boldsymbol{A}^3)$.

Solution

(a).

$$\det(\boldsymbol{AB}) = \det(\boldsymbol{A})\det(\boldsymbol{B}) = 12.$$

(b).

$$\det(5\boldsymbol{A}) = 5^3 \det(\boldsymbol{A}) = -375.$$

(c).

$$\det(\boldsymbol{B}^T) = \det(\boldsymbol{B}) = -4.$$

(d).

$$\det(\boldsymbol{A}^{-1}) = \frac{1}{\det(\boldsymbol{A})} = -\frac{1}{3}.$$

(e).

$$\det(\boldsymbol{A}^3) = (\det(\boldsymbol{A}))^3 = -27.$$

Question 6. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be 2×2 matrices and let

$$\boldsymbol{C} = \begin{bmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{bmatrix}, \qquad \boldsymbol{D} = \begin{bmatrix} b_{11} & b_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Show that

$$\det(\boldsymbol{A} + \boldsymbol{B}) = \det(\boldsymbol{A}) + \det(\boldsymbol{B}) + \det(\boldsymbol{C}) + \det(\boldsymbol{D}).$$

Solution

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad \boldsymbol{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix},$$

$$\boldsymbol{A} + \boldsymbol{B} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix}.$$

$\det(\boldsymbol{A} + \boldsymbol{B})$
$= (a_{11} + b_{11})(a_{22} + b_{22}) - (a_{12} + b_{12})(a_{21} + b_{21})$
$= a_{11}a_{22} + a_{11}b_{22} + b_{11}a_{22} + b_{11}b_{22} - a_{21}a_{12} - a_{21}b_{12} - b_{21}a_{12} - b_{21}b_{12}$
$= (a_{11}a_{22} - a_{21}a_{12}) + (b_{11}b_{22} - b_{21}b_{12}) + (a_{11}b_{22} - b_{21}a_{12}) + (b_{11}a_{22} - a_{21}b_{12})$
$= \det(\boldsymbol{A}) + \det(\boldsymbol{B}) + \det(\boldsymbol{C}) + \det(\boldsymbol{D})$.

Question 7. Let U be a square matrix such that $U^T U = I$. Show that $\det(U) = \pm 1$.

Solution

$\det(U^T U) = \det(U^T)\det(U) = (\det(U))^2 = 1$.
So, $\det(U) = 1$ or -1.

Question 8. Assume matrix A is square, make each statement True or False

(a) If the columns of A are linearly dependent, then $\det(A) = 0$.

(b) $\det(A^{-1}) = (-1)\det(A)$.

(c) The determinant of A is the product of the diagonal entries in A.

(d) If $\det(A)$ is zero, then two rows or two columns are the same, or a row or a column is zero.

(e) $\det(A + B) = \det(A) + \det(B)$.

Solution

(a) True (b) False (c) False (d)False (e) False

Question 9. Let

$$A = \begin{bmatrix} 1 & 5 & 2 \\ 2 & 4 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

(a) Find the values of $\det(M_{21})$, $\det(M_{22})$ and $\det(M_{23})$.

(b) Find the values of A_{21}, A_{22} and A_{23}.

(c) Use your answers from part (b) to compute $\det(A)$.

Solution

(a) $\det(M_{21}) = 9,\ \det(M_{22}) = -1,\ \det(M_{23}) = -7$

(b) $A_{21} = -9,\ A_{22} = -1,\ A_{23} = 7$

(c) $\det(A) = 2A_{21} + 4A_{22} - A_{23} = -29$

Question 10. Compute determinants using the cofactor formula.

(a) $\boldsymbol{A} = \begin{bmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{bmatrix}$ (b) $\boldsymbol{B} = \begin{bmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 2 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{bmatrix}$

Solution

Find the rows or columns with many zeros then use cofactor formula. $\det(\boldsymbol{A}) = 6, \det(\boldsymbol{B}) = 6$

Question 11. Let $L_1 : \mathcal{U} \to \mathcal{V}$ and $L_2 : \mathcal{V} \to \mathcal{W}$ be linear transformations, and let $L = L_2 \circ L_2$ be the mapping defined by

$$L(\boldsymbol{u}) = L_2(L_1(\boldsymbol{u}))$$

for each $\boldsymbol{u} \in \mathcal{U}$. Show that L is a linear transformation mapping $\mathcal{U}$ into $\mathcal{W}$.

Solution

If $\boldsymbol{u}_1, \boldsymbol{u}_2 \in \mathcal{U}$, then

$$\begin{aligned} L(\alpha\boldsymbol{u}_1 + \beta\boldsymbol{u}_2) &= L_2(L_1(\alpha\boldsymbol{u}_1 + \beta\boldsymbol{u}_2)) \\ &= L_2(\alpha L_1(\boldsymbol{u}_1) + \beta L_1(\boldsymbol{u}_2)) \\ &= \alpha L_2(L_1(\boldsymbol{u}_1)) + \beta L_2(L_1(\boldsymbol{u}_2)) \\ &= \alpha L(\boldsymbol{u}_1) + \beta L(\boldsymbol{u}_2). \end{aligned}$$

Therefore L is a linear transformation.

Question 12. Consider the 3×3 Vandermonde matrix

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix}$$

(a) Show that $\det(V) = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$.

(b) What conditions must the scalars x_1, x_2 and x_3 satisfy in order for V to be nonsingular?

Solution

(a).

$\det(A) = (x_2x_3^2 - x_2^2x_3) - (x_1x_3^2 - x_1^2x_3) + (x_1x_2^2 - x_1^2x_2)$

$= x_3^2(x_2 - x_1) + x_1x_2(x_2 - x_1) + x_1x_3(x_1 - x_2) + x_2x_3(x_1 - x_2)$

$= (x_2 - x_1)(x_3^2 + x_1x_2 - x_1x_3 - x_2x_3)$

$= (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$

$(b).(x_2 - x_1)(x_3 - x_1)(x_3 - x_2) \neq 0$

that is, $x_2 \neq x_1 \text{and} x_3 \neq x_1 \text{and} x_2 \neq x_3$

Question 13. Let

$$\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \boldsymbol{b}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad \boldsymbol{b}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix},$$

and let L be the linear transformation from $\mathbb{R}_2$ into $\mathbb{R}_3$ defined by

$$L(\boldsymbol{x}) = x_1\boldsymbol{b}_1 + x_2\boldsymbol{b}_2 + (x_1 + x_2)\boldsymbol{b}_3.$$

Find the matrix $\boldsymbol{A}$ representing L with respect to the standard basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2\}$ of $\mathbb{R}_2$ and the basis $\{\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3\}$ of $\mathbb{R}_3$.

Solution

$$[L(\boldsymbol{e}_1)]_{\mathcal{B}} = \boldsymbol{b}_1 + \boldsymbol{b}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad [L(\boldsymbol{e}_2)]_{\mathcal{B}} = \boldsymbol{b}_2 + \boldsymbol{b}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Question 14. Matrix A is called to be similar to matrix B if there is an invertible matrix S such that $A = S^{-1}BS$. Now prove the following properties of similarity:

(a) Any square matrix A is similar to itself.

(b) If B is similar to A, then A is similar to B.

(c) If A is similar to B and B is similar to C, then A is similar to C.

Solution

(a)

$$A = I^{-1}AI$$

(b) If $B = S^{-1}AS$ then $A = SBS^{-1} = (S^{-1})^{-1}BS^{-1} = S_1^{-1}BS_1$, where $S_1 = S^{-1}$.

(c) If $A = S^{-1}BS$ and $B = T^{-1}CT$ then

$$A = S^{-1}\left(T^{-1}CT\right)S = \left(S^{-1}T^{-1}\right)C(TS) = (TS)^{-1}C(TS) = S_2^{-1}CS_2$$

where $S_2 = TS$

Question 15. Find the matrix representation of the linear transformation $T(f(t)) = f(3t-2)$ from $\mathbb{P}_2$ to $\mathbb{P}_2$, where $\mathbb{P}_2 = \{a_0 + a_1x + a_2x^2 \mid a_0, a_1, a_2 \in \mathbb{R}\}$.

Solution

A basis for P_2 is : $[1, t, t^2]$.

$$\begin{aligned} T(1) &= 1 = 1\cdot 1 + 0\cdot t + 0\cdot t^2 \\ T(t) &= 3t - 2 = -2\cdot 1 + 3\cdot t + 0\cdot t^2 \\ T\left(t^2\right) &= (3t-2)^2 = 4\cdot 0 - 12\cdot t + 9t^2 \end{aligned}$$

the matrix of T w.r.t to P_2 is

$$B = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 3 & -12 \\ 0 & 0 & 9 \end{bmatrix}$$

Question 16. Let L be the linear tranformation mapping $\mathbb{R}^3$ into $\mathbb{R}^3$ defined by $L(x) = Ax$, where

$$A = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$$

and let

$$\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \boldsymbol{v}_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

Find the transition matrix V corresponding to a change of basis from $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ to $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$, and use it to determine the matrix B representing L with respect to $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$.

Solution

$$V = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -2 \\ 1 & 0 & 1 \end{bmatrix}. \quad V^{-1} = \begin{bmatrix} -2 & 1 & 2 \\ 3 & -1 & -2 \\ 2 & -1 & -1 \end{bmatrix}$$

$$B = V^{-1}AV = \begin{bmatrix} -2 & 1 & 2 \\ 3 & -1 & -2 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -2 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 17. A linear transformation $L : V \to W$ is said to be one-to-one if $L(\mathbf{v}_1) = L(\mathbf{v}_2)$ implies that $\mathbf{v}_1 = \mathbf{v}_2$ (i.e., no two distinct vectors $\mathbf{v}_1, \mathbf{v}_2$ in V get mapped into the same vector $\mathbf{w} \in W$. Show that L is one-to-one if and only if $\ker(L) = \{0_V\}$.

Solution

Suppose L is ore-to-one and $\underline{v} \in \ker(L)$.

$$L(\underline{v}) = 0_w \quad \text{and} \quad L(0_v) = 0_w$$

Since L is one-to-one, it follows that $\underline{v} = 0_v$, Therefore $\ker(L) = \{0_v\}$.
Conversely, suppose $\ker(L) = \{0_v\}$ and $L(v_1) = L(v_2)$. Then

$$L(v_1 - v_2) = L(v_1) - L(v_2) = 0_w$$

Therefore $v_1 - v_2 \in \ker(L)$ and here

$$v_1 - v_2 = 0_v$$
$$v_1 = v_2$$

Since L is one-to-one.

Question 18. For the following linear transformations $T : \mathbb{R}^n \to \mathbb{R}^n$, find a matrix A such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^3$,

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x - y \\ 3y \\ 4x + 5y \end{bmatrix}$$

(b) $T : \mathbb{R}^2 \to \mathbb{R}^2$, satisfying

$$T\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, T\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

Solution

(a) To figure out the matrix for a linear transformation from $\mathbb{R}^n$, we find the matrix A whose first column is $T(\vec{e}_1)$, whose second column is $T(\vec{e}_2)$ - in general, whose i th column is $T(\vec{e}_i)$. Here, by definition we have that

$$T(\vec{e}_1) = T\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}, T(\vec{e}_2) = T\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 5 \end{bmatrix}$$

Thus,

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 3 \\ 4 & 5 \end{bmatrix}$$

(b) We need to find $T(\vec{e}_2)$ and $T(\vec{e}_2)$. Given the information we have, this is easiest to do by writing $\vec{e}_1$ and $\vec{e}_2$ as linear combinations of

$$\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$$

We start with $\vec{e}_1$. We solve

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Setting up the system of equations as usual and solving yields $c_1 = 3, c_2 = -1$. Thus, we have that

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Now, since $T(\vec{x} + \vec{y}) = T(\vec{x}) + T(\vec{y})$, and $T(c\vec{x}) = cT(\vec{x})$, this gets us that

$$\begin{aligned} T\begin{bmatrix} 1 \\ 0 \end{bmatrix} &= T\left(3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) \\ &= T\left(3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) + T\left((-1) \begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) \\ &= 3T \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-1)T \begin{bmatrix} 2 \\ 3 \end{bmatrix} \\ &= 3 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + (-1) \begin{bmatrix} -2 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ -11 \end{bmatrix} \end{aligned}$$

Similarly, we get that

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = (-2)\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

and a calculation like the one above yields

$$\begin{aligned} T\begin{bmatrix} 0 \\ 1 \end{bmatrix} &= (-2)T\begin{bmatrix} 1 \\ 1 \end{bmatrix} + T\begin{bmatrix} 2 \\ 3 \end{bmatrix} \\ &= (-2)\begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} -2 \\ 5 \end{bmatrix} = \begin{bmatrix} -4 \\ 9 \end{bmatrix} \end{aligned}$$

Combining the information, we see that

$$A = \begin{bmatrix} 5 & -4 \\ -11 & 9 \end{bmatrix}$$

Question 19. Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \vec{v}_1 = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$, and $\vec{v}_2 = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$, and let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation that maps $\vec{x}$ into $x_1\vec{v}_1 + x_2\vec{v}_2$. Find a matrix A such that $T(\vec{x})$ is $A\vec{x}$ for each $\vec{x}$.

Solution

We have

$$A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) \end{bmatrix} = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 \end{bmatrix} = \begin{bmatrix} -3 & 7 \\ 5 & -2 \end{bmatrix}$$

Question 20. Show that the transformation T defined by

$$T(x_1, x_2) = (x_1 - 2|x_2|, x_1 - 4x_2)$$

is not linear.

Solution

We have

$$T(0,1) + T(0,-1) = (-2,-4) + (-2,4) = (-4,0)$$

but

$$T((0,1) + (0,-1)) = T(0,0) = (0,0)$$

These are not equal, so T does not preserve vector addition. Therefore it is not a linear transformation.